2018 H2 Math 9758 Paper 2 Solution

	Solution
1(i)	$$\frac{dy}{dx}=\left(\frac{1}{3}y-15\right)^{\frac{1}{3}}$$ $$\left(\frac{1}{3}y-15\right)^{-\frac{1}{3}}\frac{dy}{dx}=1$$ $$\int\left(\frac{1}{3}y-15\right)^{-\frac{1}{3}}\,dy=\int 1\,dx$$ $$\frac{\left(\frac{1}{3}y-15\right)^{\frac{2}{3}}}{\frac{2}{3}\left(\frac{1}{3}\right)}=x+C$$ $$\frac{9}{2}\left(\frac{1}{3}y-15\right)^{\frac{2}{3}}=x+C\quad ---(1)$$ Apply Variable separable. $\left(\frac{1}{3}y-15\right)^{-\frac{1}{3}}$ must integrate w.r.t. *y*. Since $\frac{1}{3}y-15$ is linear, we can integrate directly and we have to remember to divide by the coefficient of *y*. We are applying: $$\int y^n\,dy=\frac{y^{n+1}}{n+1}+c\quad ,n\neq-1$$ $$\int(ay+b)^n\,dy=\frac{1}{a}\left(\frac{(ay+b)^{n+1}}{n+1}\right)+c\quad ,n\neq-1$$ When $x=0, y=69$, $$\frac{9}{2}\left(\frac{1}{3}(69)-15\right)^{\frac{2}{3}}=0+C$$ $$18=C$$ $$C=18$$ Since there is no modulus involved in your equation, you can sub in the given conditions to find your arbitrary constant *C*. Sub $C=18$ into (1) $$\frac{9}{2}\left(\frac{1}{3}y-15\right)^{\frac{2}{3}}=x+18\quad ---(2)$$ $$\left(\frac{1}{3}y-15\right)^{\frac{2}{3}}=\frac{2}{9}x+4$$ $$\frac{1}{3}y-15=\left(\frac{2}{9}x+4\right)^{\frac{3}{2}}$$ $$\frac{1}{3}y=\left(\frac{2}{9}x+4\right)^{\frac{3}{2}}+15$$ $$y=3\left(\frac{2}{9}x+4\right)^{\frac{3}{2}}+45$$ $$\therefore f(x)=3\left(\frac{2}{9}x+4\right)^{\frac{3}{2}}+45$$ Answer the question.

(ii)

Sub $\frac{dy}{dx}=4$ into $\frac{dy}{dx}=\left(\frac{1}{3}y-15\right)^{\frac{1}{3}}$

$$\left(\frac{1}{3}y-15\right)^{\frac{1}{3}}=4$$
$$\frac{1}{3}y-15=64$$
$$\frac{1}{3}y=79$$
$$y=237$$

Sub $y=237$ into (2)

$$\frac{9}{2}\left(\frac{1}{3}(237)-15\right)^{\frac{2}{3}}=x+18$$
$$x+18=72$$
$$x=54$$

Sub into the equation where x is the subject for ease of manipulation.

Therefore, when the gradient is 4, the coordinates are $(54, 237)$.

2(i)

<u>Method 1 (Use of GC allowed):</u>

Since $2-3\mathrm{i}$ is a root, sub $2-3\mathrm{i}$ into the given equation:

$$4(2-3\mathrm{i})^4-20(2-3\mathrm{i})^3+s(2-3\mathrm{i})^2-56(2-3\mathrm{i})+t=0$$
$$332+828\mathrm{i}+s(-5-12\mathrm{i})+t=0$$
$$332+828\mathrm{i}-5s-12s\mathrm{i}+t=0$$
$$(332-5s+t)+(828-12s)\mathrm{i}=0+0\mathrm{i}$$

Use GC to simplify the expression.

Comparing real and imaginary components,

Imaginary: $828-12s=0 \quad\Rightarrow\quad s=69$
Real: $332-5(69)+t=0 \quad\Rightarrow\quad t=13$

$$4x^4-20x^3+69x^2-56x+13=0$$

Use GC (APPS: 4:Plysmlt2 / 1: POLYNOMIAL ROOT FINDER)

Using GC, $x=2+3\mathrm{i}$ or $x=2-3\mathrm{i}$ or $x=\frac{1}{2}$.

Therefore, the other roots are $2+3\mathrm{i}$ and $\frac{1}{2}$. Also, $s=69$ and $t=13$.

	Method 2 (Use of GC NOT allowed): Since the equation has real coefficients and $2-3\mathrm{i}$ is a root, then $2+3\mathrm{i}$ is also a root. $\text{Quadratic Factor} = \left(x-(2-3\mathrm{i})\right)\left(x-(2+3\mathrm{i})\right)$ $= \left((x-2)+3\mathrm{i}\right)\left((x-2)-3\mathrm{i}\right)$ $= (x-2)^2-(3\mathrm{i})^2$ $= x^2-4x+4+9$ $= x^2-4x+13$ $4x^4-20x^3+sx^2-56x+t = (x^2-4x+13)\left(4x^2+ax+\frac{t}{13}\right)$ where $a \in \mathbb{R}$ Comparing coefficients of $x^3: -20=-16+a$ $a=-4$ $x: -56=-4\left(\frac{t}{13}\right)-4(13)$ $t=13$ $x^2: s=1-4(-4)+13(4)=69$ $4x^4-20x^3+69x^2-56x+t=0$ $(x^2-4x+13)(4x^2-4x+1)=0$ $(x^2-4x+13)(2x-1)^2=0$ $x=\frac{1}{2}$ (repeated roots) or $x=2\pm3\mathrm{i}$ Therefore, the other roots are $2+3\mathrm{i}$ and $\frac{1}{2}$. Also, $s=69$ and $t=13$.
(bi)	**Method 1 (Recommended):** Since $w=3$ is a root, $(w-3)$ is a factor of the equation. $w^3-27=(w-3)(w^2+aw+9)$ where $a \in \mathbb{R}$ Comparing coefficient of w: $0=-3a+9 \Rightarrow a=3$ $w^2+3w+9=0$ $w=\frac{-3\pm\sqrt{3^2-4(9)}}{2}$ $=-\frac{3}{2}\pm\frac{3\sqrt{3}}{2}\mathrm{i}$

> Give your reason clearly.

> Rearrange the expression such that it becomes the form $(a-b)(a+b)=a^2-b^2$ for ease of expansion.

> With one quadratic factor, we can proceed to factorise the given equation to solve for the unknowns (by comparing coefficients) and it will, in the process, solve for the roots since we have factorised the equation

> With one linear factor, we can proceed to factorise the given equation to solve for the roots by first finding the other quadratic factor.

> To solve quadratic equations, we have the 4 methods:
> 1) Use calculator (not allowed in this question)
> 2) Factorisation
> 3) Complete the square
> 4) Use formula

Therefore, the other possible values of w are $-\frac{3}{2}+\frac{3\sqrt{3}}{2}\mathrm{i}$ and $-\frac{3}{2}-\frac{3\sqrt{3}}{2}\mathrm{i}$.

Method 2 (Not recommended):

Let $w = a + b\mathrm{i}$ where $a, b \in \mathbb{R}$.

$$w^3 = 27$$

$$(a+b\mathrm{i})^3 = 27$$

$$a^3 + 3a^2b\mathrm{i} - 3ab^2 - b^3\mathrm{i} = 27$$

Comparing real and imaginary components,

$$a^3 + 3a^2b\mathrm{i} - 3ab^2 - b^3\mathrm{i} = 27$$

$$a^3 - 3ab^2 = 27 \text{ ------ } (1)$$

$$3a^2b - b^3 = 0$$

$$b\left(3a^2 - b^2\right) = 0$$

$b = 0$ or $b^2 = 3a^2$

Substituting into (1),

$a^3 = 27$ or $a^3 - 9a^3 = 27$

$a = 3$ $\qquad a = -\frac{3}{2}$

$$b^2 = \frac{27}{4}$$

$$b = \pm\frac{3\sqrt{3}}{2}$$

Therefore, the other possible values of w are $-\frac{3}{2}+\frac{3\sqrt{3}}{2}\mathrm{i}$ and $-\frac{3}{2}-\frac{3\sqrt{3}}{2}\mathrm{i}$.

Method 3 (for Further Math students):

$$w^3 = 27$$

$$= 27\mathrm{e}^{\mathrm{i}(0+2k\pi)}$$

$$w = 3\mathrm{e}^{\mathrm{i}\frac{2}{3}k\pi}, \quad k = -1, 0, 1$$

$w = 3\mathrm{e}^{\mathrm{i}\left(-\frac{2}{3}\pi\right)} = 3\left(\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}\right)$ or $w = 3\mathrm{e}^{\mathrm{i}0} = 3$ or $w = 3\mathrm{e}^{\mathrm{i}\left(\frac{2}{3}\pi\right)} = 3\left(\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}\right)$

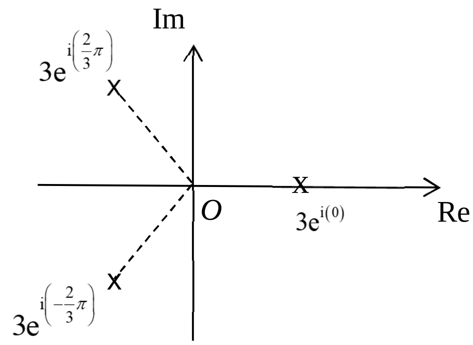
Therefore, the other possible values of w are $-\frac{3}{2}+\frac{3\sqrt{3}}{2}\mathrm{i}$ and $-\frac{3}{2}-\frac{3\sqrt{3}}{2}\mathrm{i}$.

(ii) Using GC,

$$w=-\frac{3}{2}+\frac{3\sqrt{3}}{2}\mathrm{i} = 3\mathrm{e}^{\mathrm{i}\left(\frac{2}{3}\pi\right)}$$

$$w=-\frac{3}{2}-\frac{3\sqrt{3}}{2}\mathrm{i} = 3\mathrm{e}^{\mathrm{i}\left(-\frac{2}{3}\pi\right)}$$

$$w=3 = 3\mathrm{e}^{\mathrm{i}(0)}$$



(iii)

$$\text{Sum}=3+\left(-\frac{3}{2}+\frac{3\sqrt{3}}{2}\mathrm{i}\right)+\left(-\frac{3}{2}-\frac{3\sqrt{3}}{2}\mathrm{i}\right) = 0$$

For addition of complex numbers, use **Cartesian Form** where possible.

$$\text{Product}=3\left(3\mathrm{e}^{\mathrm{i}\left(\frac{2}{3}\pi\right)}\right)\left(3\mathrm{e}^{\mathrm{i}\left(-\frac{2}{3}\pi\right)}\right) = 27$$

For multiplication of complex numbers, use **Exponential Form** where possible.

3(i)

$$\overrightarrow{AD}=\overrightarrow{BC}$$

$$\overrightarrow{OD}-\overrightarrow{OA}=\overrightarrow{OC}-\overrightarrow{OB}$$

$$\overrightarrow{OD}-\begin{pmatrix}5\\-4\\1\end{pmatrix}=\begin{pmatrix}-5\\4\\2\end{pmatrix}-\begin{pmatrix}5\\4\\0\end{pmatrix}$$

$$\overrightarrow{OD}=\begin{pmatrix}-5\\-4\\3\end{pmatrix}$$

Since *ABCD* forms a parallelogram, then $\overrightarrow{AD}$ and $\overrightarrow{BC}$ are parallel and equal in length (and in the same direction).

We can also say the same for $\overrightarrow{AB}$ and $\overrightarrow{DC}$.

Therefore, coordinates of *D* are $(-5,-4,3)$.

Answer the question. Give your answer in coordinates form.

(ii)

$$\overrightarrow{BE} = \overrightarrow{OE} - \overrightarrow{OB} = \begin{pmatrix} 0 \\ 0 \\ 10 \end{pmatrix} - \begin{pmatrix} 5 \\ 4 \\ 0 \end{pmatrix} = \begin{pmatrix} -5 \\ -4 \\ 10 \end{pmatrix}$$

$$\overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{OB} = \begin{pmatrix} -5 \\ 4 \\ 2 \end{pmatrix} - \begin{pmatrix} 5 \\ 4 \\ 0 \end{pmatrix} = \begin{pmatrix} -10 \\ 0 \\ 2 \end{pmatrix}$$

$$\overrightarrow{BE} \times \overrightarrow{BC} = \begin{pmatrix} -5 \\ -4 \\ 10 \end{pmatrix} \times \begin{pmatrix} -10 \\ 0 \\ 2 \end{pmatrix}$$

To get the normal vector of the place BCE, we cross 2 non-parallel vectors ($\overrightarrow{BE}$ and $\overrightarrow{BC}$) that are parallel to the plane.

$$= \begin{pmatrix} -8 \\ -90 \\ -40 \end{pmatrix} = -2 \begin{pmatrix} 4 \\ 45 \\ 20 \end{pmatrix}$$

$$\mathbf{r} \bullet \begin{pmatrix} 4 \\ 45 \\ 20 \end{pmatrix} = \begin{pmatrix} 4 \\ 45 \\ 20 \end{pmatrix} \bullet \begin{pmatrix} 0 \\ 0 \\ 10 \end{pmatrix}$$

$$= 200$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \bullet \begin{pmatrix} 4 \\ 45 \\ 20 \end{pmatrix} = 200$$

$$4x + 45y + 20z = 200$$

Answer the question. Give your answer in **Cartesian form**.

(iii)

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 5 \\ 4 \\ 0 \end{pmatrix} - \begin{pmatrix} 5 \\ -4 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 8 \\ -1 \end{pmatrix}$$

$$\overrightarrow{AB} \times \overrightarrow{BC} = \begin{pmatrix} 0 \\ 8 \\ -1 \end{pmatrix} \times \begin{pmatrix} -10 \\ 0 \\ 2 \end{pmatrix}$$

To get the normal vector of the place $ABCD$, we cross 2 non-parallel vectors ($\overrightarrow{AB}$ and $\overrightarrow{BC}$) that are parallel to the plane.

$$= \begin{pmatrix} 16 \\ 10 \\ 80 \end{pmatrix} = 2 \begin{pmatrix} 8 \\ 5 \\ 40 \end{pmatrix}$$

For **acute angle** (for between 2 planes, 2 lines, or a line and a plane), always input the **modulus**.

$$\text{Required angle} = \cos^{-1} \frac{\left| \begin{pmatrix} 8 \\ 5 \\ 40 \end{pmatrix} \bullet \begin{pmatrix} 4 \\ 45 \\ 20 \end{pmatrix} \right|}{\sqrt{1689}\sqrt{2441}}$$

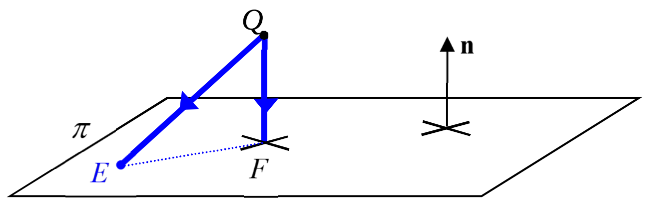
$$= 58.6^\circ \quad (1 \text{ d.p.})$$

Always give **1 decimal place for angles in degree** unless exact or specified otherwise by question.

(iv) Let Q be the midpoint of AD

$$\overrightarrow{OQ}=\frac{\begin{pmatrix}5\\-4\\1\end{pmatrix}+\begin{pmatrix}-5\\-4\\3\end{pmatrix}}{2}=\begin{pmatrix}0\\-4\\2\end{pmatrix}$$

Apply Ratio Theorem or in this case, Midpoint Theorem.



Shortest distance $= QF$

$= \text{length of projection of } \overrightarrow{QE} \text{ onto } \mathbf{n}$

$$=\frac{\left|\overrightarrow{QE}\bullet\mathbf{n}\right|}{|\mathbf{n}|}$$

Remember your modulus.

$$=\frac{\left|\left(\begin{pmatrix}0\\0\\10\end{pmatrix}-\begin{pmatrix}0\\-4\\2\end{pmatrix}\right)\bullet\begin{pmatrix}4\\45\\20\end{pmatrix}\right|}{\sqrt{2441}}$$

$= 6.88 \quad (3 \text{ s.f.})$

4(i)

From MF26:

$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{4!} - \ldots + \frac{(-1)^r x^{2r}}{(2r)!} + \ldots$$

Replace x by $2x$

And we need to expand until $r = 3$ to get x^6.

$$\ln(\cos(2x))$$

$$= \ln\left(1 - \frac{(2x)^2}{2} + \frac{(2x)^4}{4!} - \frac{(2x)^6}{6!} + \ldots\right)$$

$$= \ln\left(1 + \left(-\frac{(2x)^2}{2} + \frac{(2x)^4}{4!} - \frac{(2x)^6}{6!} + \ldots\right)\right)$$

$$= \left(-\frac{(2x)^2}{2} + \frac{(2x)^4}{4!} - \frac{(2x)^6}{6!} + ..\right) - \frac{\left(-\frac{(2x)^2}{2} + \frac{(2x)^4}{4!} - \frac{(2x)^6}{6!} + ..\right)^2}{2}$$

$$+ \frac{\left(-\frac{(2x)^2}{2} + \frac{(2x)^4}{4!} - \frac{(2x)^6}{6!} + ..\right)^3}{3} + \ldots$$

From MF26:

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \ldots + \frac{(-1)^{r+1} x^{2r}}{r} + \ldots \quad (-1 < x \le 1)$$

Replace x by $\left(-\frac{(2x)^2}{2} + \frac{(2x)^4}{4!} - \frac{(2x)^6}{6!} + \ldots\right)$

We require only the first 3 terms to get x^6.

$$= \left(-2x^2 + \frac{2x^4}{3} - \frac{4x^6}{45}\right) - \frac{1}{2}\left(\left(-\frac{(2x)^2}{2}\right)^2 + 2\left(-\frac{(2x)^2}{2}\right)\left(\frac{(2x)^4}{4!}\right)\right) + \frac{1}{3}\left(-\frac{(2x)^2}{2}\right)^3 + \ldots$$

$$= \left(-2x^2 + \frac{2x^4}{3} - \frac{4x^6}{45}\right) - \frac{1}{2}\left(4x^4 - \frac{8x^6}{3}\right) + \frac{1}{3}\left(-8x^6\right) + \ldots$$

$$= -2x^2 + \frac{2x^4}{3} - \frac{4x^6}{45} - 2x^4 + \frac{4x^6}{3} - \frac{8x^6}{3} + \ldots$$

$$= -2x^2 - \frac{4}{3}x^4 - \frac{64}{45}x^6 + \ldots \quad \left(\text{up to } x^6\right)$$

	The expansion is not valid when $\ln(\cos(2x))$ is undefined. That means $\cos(2x)=0$ Thus, the expansion is not valid when $x=\frac{\pi}{4}$. **<u>Method 2</u>** $y=\ln(\cos(2x))$ $\frac{dy}{dx}=-2\tan(2x)$ $\frac{d^2y}{dx^2}=-4\sec^2(2x)$ $\frac{d^3y}{dx^3}=-16\sec^2(2x)\tan(2x)$ $\frac{d^4y}{dx^4}=-64\sec^2(2x)\tan^2(2x)+32\sec^4(2x)$ Using GC, when $x=0$, $y=0$ $\frac{dy}{dx}=0$ $\frac{d^2y}{dx^2}=-4$ $\frac{d^3y}{dx^3}=0$ $\frac{d^4y}{dx^4}=-32$ $\frac{d^5y}{dx^5}=0$ $\frac{d^6y}{dx^6}=-1024$ With the 4th derivative, we can use GC to get the values of the 5th and 6th derivatives when $x=0$. **Note**: GC can only generate up to 2 derivatives from input thus we still need to find up to the 4th derivative in order to get the 6th derivative. $\ln(\cos(2x))$ $=-\frac{4}{2}x^2-\frac{16}{4!}x^4-\frac{512}{6!}x^6+\ldots$ $=-2x^2-\frac{4}{3}x^4-\frac{64}{45}x^6+\ldots \quad (\text{up to } x^6)$ The expansion is not valid when $x=\frac{\pi}{4}$.
(ii)	$\int \frac{\ln(\cos(2x))}{x^2}\,dx \approx \int \frac{-2x^2-\frac{4}{3}x^4-\frac{64}{45}x^6}{x^2}\,dx \quad (\text{up to } x^6\text{, from (i)})$ $=\int -2-\frac{4}{3}x^2-\frac{64}{45}x^4\,dx$ $=-2x-\frac{4}{9}x^3-\frac{64}{225}x^5+C$ $\int_0^{0.5} \frac{\ln(\cos(2x))}{x^2}\,dx \approx \left[-2x-\frac{4}{9}x^3-\frac{64}{225}x^5\right]_0^{0.5}$ $=-1.0644 \quad (4\text{ d.p.})$

(iii) $\int_0^{0.5} \frac{\ln(\cos(2x))}{x^2}\,\mathrm{d}x = -1.0670 \quad (4\text{ d.p.})$

5(i) A sample of 30 fans is sufficient for Central Limit to apply as we would require the sample statistic, in this case, the sample mean to be normally distributed for the test to be valid. Thus, there is no need to know the distribution of the times of failures for the fans.

The fans should be chosen randomly, that is, each fan has an equal chance of being chosen, and the fans should be chosen independently too.

(ii) Let μ be the population MTTF (in hours).

Null hypothesis, $H_0 : \mu = 65000$

Alternative hypothesis, $H_1 : \mu < 65000$

(iii) Let X be the MTTF (in hours).

Let s^2 be the variance used in calculating the test statistics (in hours^2).

Under H_0 ,since sample size $n = 43$ is sufficiently large, by Central Limit Theorem,

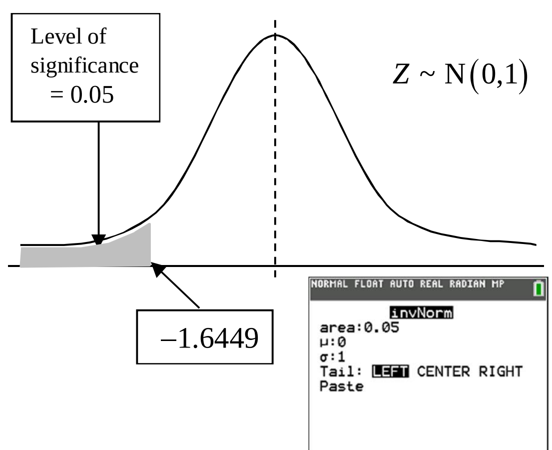
$\overline{X} \sim N\left(65000, \frac{s^2}{43}\right)$ approximately.

Test statistics: $Z = \dfrac{\overline{X} - 65000}{s/\sqrt{43}}$

Level of significance: 5%

Reject H_0 if $z-\text{value} < -1.6449$

$z-\text{value} = \dfrac{64230 - 65000}{s/\sqrt{43}}$



Since the null hypothesis is not rejected at 5% level of significance,

$$z-\text{value} > -1.6449$$

$$\frac{64230 - 65000}{s/\sqrt{43}} > -1.6449$$

$$-770\sqrt{43} > -1.6449s$$

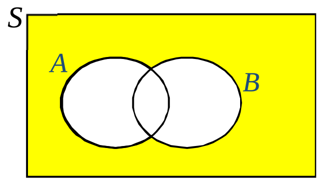
$$s > 3069.7$$

$$s^2 > 9.42 \times 10^6 \quad (3\text{ s.f.})$$

There is an unknown in s^2 so we have to use the z-value method.

6(i)	To end up at D, the bug must take exactly 5 left forks and 3 right forks, out of a total of 8 junctions and this can be done by ${}^{8}C_{5}$ ways.. Required probability $= {}^{8}C_{5} \times p \times p \times p \times p \times p \times q \times q \times q$ $= {}^{8}C_{5}\, p^{5} q^{3}$ $= 56 p^{5} q^{3} \quad \text{(Shown)}$ We are expected to explain clearly the number of left and right forks the bug has to take to get ${}^{8}C_{5}$.We can also use ${}^{8}C_{3}$.
(ii)	Let X be the number of left forks taken by the bug. $X \sim \mathrm{B}(8, p)$ Since the bug has greater probability of finishing its journey at D than at any other endpoints, Same meaning as $X = 5$ is the mode. For binomial distribution, if $X = 5$ is the mode then we will get (i) $\mathrm{P}(X=0) < \mathrm{P}(X=1) < \mathrm{P}(X=2) < \mathrm{P}(X=3) < \mathrm{P}(X=4) < \mathrm{P}(X=5)$ (ii) $\mathrm{P}(X=5) > \mathrm{P}(X=6) > \mathrm{P}(X=7) > \mathrm{P}(X=8)$ Thus, we only need $\mathrm{P}(X=4) < \mathrm{P}(X=5)$ **and** $\mathrm{P}(X=5) > \mathrm{P}(X=6)$. $\mathrm{P}(X=4) < \mathrm{P}(X=5)$ $70 p^{4} q^{4} < 56 p^{5} q^{3}$ $70 p^{4}(1-p)^{4} < 56 p^{5}(1-p)^{3}$ $5(1-p) < 4p$ $p > \frac{5}{9}$ and $\mathrm{P}(X=5) > \mathrm{P}(X=6)$ $56 p^{5} q^{3} > 28 p^{6} q^{2}$ $56 p^{5}(1-p)^{3} > 28 p^{6}(1-p)^{2}$ $2(1-p) > p \quad (\text{since } p, q > 0)$ $p < \frac{2}{3}$ "and" means "intersection" $\therefore \frac{5}{9} < p < \frac{2}{3}$ Exactly: exact answers are expected.
(iii)	Required probability $= (1-0.1)^{8}$ $= 0.43046721$ $= 0.430 \quad (3 \text{ s.f.})$ At each junction, the probability of not being swallowed up by a black hole at each junction $= 1-0.1$. Thus, to reach any one of the end points, the bug must not be swallowed up at all 8 junctions.

7(i)



$$\begin{aligned}\mathrm{P}(A'\cap B') &= 1-\mathrm{P}(A\cup B)\\ &= 1-\left(\mathrm{P}(A)+\mathrm{P}(B)-\mathrm{P}(A\cap B)\right)\\ &= 1-\left(\mathrm{P}(A)+\mathrm{P}(B)-\mathrm{P}(A)\times\mathrm{P}(B)\right) \quad (\text{since } A \text{ and } B \text{ independent})\\ &= 1-a-b+ab\end{aligned}$$

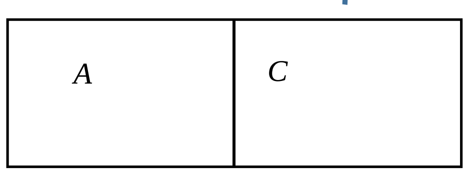
Since $\mathrm{P}(A'\cap B') = 1-a-b+ab$ and $\mathrm{P}(A')\times\mathrm{P}(B') = (1-a)(1-b) = 1-a-b+ab$, then $\mathrm{P}(A'\cap B') = \mathrm{P}(A')\times\mathrm{P}(B')$.

$\therefore A'$ and B' independent

(ii)

$$\begin{aligned}\mathrm{P}(A'\cap C') &= 1-\mathrm{P}(A\cup C)\\ &= 1-\left(\mathrm{P}(A)+\mathrm{P}(C)-\mathrm{P}(A\cap C)\right)\\ &= 1-\left(\mathrm{P}(A)+\mathrm{P}(C)-0\right) \quad (\text{since } A \text{ and } C \text{ mutually exclusive})\\ &= 1-a-c\end{aligned}$$

Since A' and C' are mutually exclusive, $\mathrm{P}(A'\cap C') = 0$, that is $\mathrm{P}(A)+\mathrm{P}(C) = 1$. Also, *A* and *C* are mutually exclusive, meaning $\mathrm{P}(A\cap C) = 0$. Thus, we split the Venn diagram into *A* and *C*.



(iii)

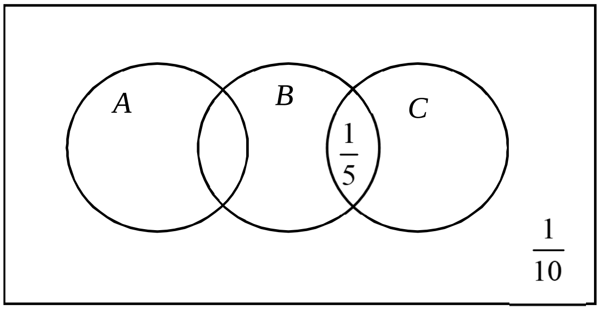
Step 1:

Given that

1) A' and C' are not mutually exclusive.
2) A and C are mutually exclusive.
3) $\mathrm{P}(B\cap C)=\frac{1}{5}$ and $\mathrm{P}(A'\cap B'\cap C')=\frac{1}{10}$

we can represent that using a Venn diagram for ease of understanding:

Using the basic information given, sketch a Venn diagram and fill in all necessary info.



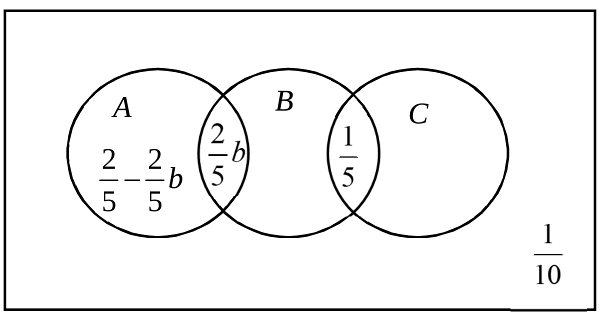
Step 2:

We are also given that $\mathrm{P}(A)=\frac{2}{5}$ and $\mathrm{P}(B)=b$

since A, B independent, $\mathrm{P}(A\cap B)=\mathrm{P}(A)\mathrm{P}(B)=\frac{2}{5}b$.

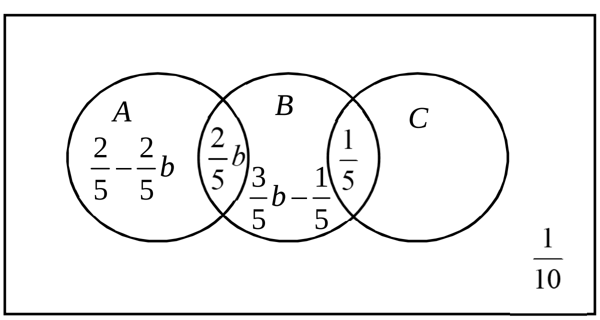
Adding these information into the Venn diagram gives us:

Using the additional information, we can fill in sections of the Venn diagram step by step.



Step 3:

Since we have $\mathrm{P}(B)=b$, we can fill $\mathrm{P}(A'\cap B\cap C')=b-\frac{2}{5}b-\frac{1}{5}=\frac{3}{5}b-\frac{1}{5}$



Using the additional information, we can fill in sections of the Venn diagram step by step.

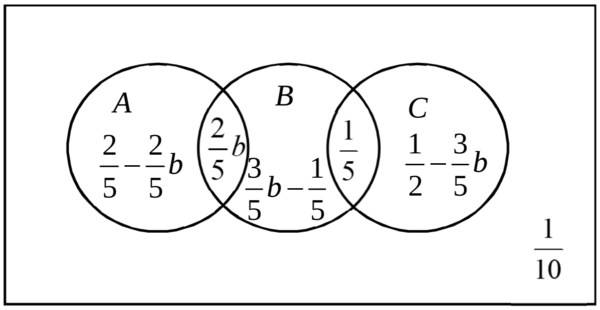
Step 4:

Since sum of all probabilities $=1$, we have

$$P(B'\cap C)=1-\left(\frac{1}{10}+\frac{2}{5}-\frac{2}{5}b+b\right)=\frac{1}{2}-\frac{3}{5}b$$

Important:

Sum of all probabilities in a Venn diagram must equal to 1.



Note that all probabilities must be greater than or equal to 0.

So we must have

$$\frac{3}{5}b-\frac{1}{5}\geq 0 \qquad b\geq\frac{1}{3}$$

and

$$\frac{1}{2}-\frac{3}{5}b\geq 0 \qquad b\leq\frac{5}{6}$$

Basic Result:

All probabilities lie between 0 and 1 (both inclusive).

Thus, $\frac{1}{3}\leq b\leq\frac{5}{6}$

Therefore $\min P(A\cap B)=\frac{2}{5}b=\frac{2}{5}\left(\frac{1}{3}\right)=\frac{2}{15}$

and $\max P(A\cap B)=\frac{2}{5}b=\frac{2}{5}\left(\frac{5}{6}\right)=\frac{1}{3}$

8(i)

$$P(S=6)=P(\text{both balls are '3'})=\frac{2}{n+5}\times\frac{1}{n+4}=\frac{2}{(n+5)(n+4)}$$

$$P(S=7)=P(\text{one '3' and one '4'})=\frac{2}{n+5}\times\frac{3}{n+4}\times 2=\frac{12}{(n+5)(n+4)}$$

$$\mathrm{P}(S=8)=\mathrm{P}(\text{one '3' and one '5'})+\mathrm{P}(\text{both balls are '4'})$$

$$=\frac{2}{n+5}\times\frac{n}{n+4}\times 2+\frac{3}{n+5}\times\frac{2}{n+4}$$

$$=\frac{4n+6}{(n+5)(n+4)}$$

$$\mathrm{P}(S=9)=\mathrm{P}(\text{one '4' and one '5'})=\frac{3}{n+5}\times\frac{n}{n+4}\times 2=\frac{6n}{(n+5)(n+4)}$$

$$\mathrm{P}(S=10)=\mathrm{P}(\text{both balls are '5'})=\frac{n}{n+5}\times\frac{n-1}{n+4}=\frac{n(n-1)}{(n+5)(n+4)}$$

s	6	7	8
$\mathrm{P}(S=s)$	$\frac{2}{(n+5)(n+4)}$	$\frac{12}{(n+5)(n+4)}$	$\frac{4n+6}{(n+5)(n+4)}$

s	9	10
$\mathrm{P}(S=s)$	$\frac{6n}{(n+5)(n+4)}$	$\frac{n(n-1)}{(n+5)(n+4)}$

(ii)

For $n=1$, $\mathrm{P}(S=10)=0$.

It is not possible to achieve 10 as at least 2 balls numbered 5 are required.

(iii)

$$\mathrm{E}(S)=\frac{6(2)+7(12)+8(4n+6)+9(6n)+10n(n-1)}{(n+5)(n+4)}$$

$$=\frac{10n^2+76+144}{(n+5)(n+4)}$$

$$=\frac{(10n+36)(n+4)}{(n+5)(n+4)}$$

$$=\frac{10n+36}{n+5} \quad (\text{shown})$$

Show every single step including the factorisation that leads to the simplification of the fraction.

$$\mathrm{E}\left(S^2\right)=\frac{6^2(2)+7^2(12)+8^2(4n+6)+9^2(6n)+10^2n(n-1)}{(n+5)(n+4)}$$

$$=\frac{100n^2+642n+1044}{(n+5)(n+4)}$$

$$\mathrm{Var}(S)=\mathrm{E}\left(S^2\right)-\left[\mathrm{E}(S)\right]^2$$

$$=\frac{100n^2+642n+1044}{(n+5)(n+4)}-\left(\frac{10n+36}{n+5}\right)^2$$

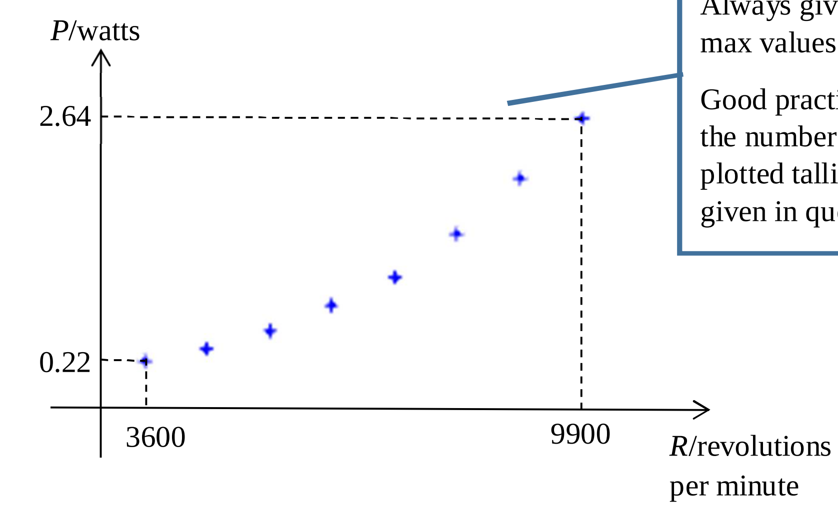
$$=\frac{\left(100n^2+642n+1044\right)(n+5)-\left(100n^2+720n+1296\right)(n+4)}{(n+5)^2(n+4)}$$

$$=\frac{22n^2+78n+36}{(n+5)^2(n+4)}$$

$$\therefore \mathrm{g}(n)=22n^2+78n+36$$

Answer the question.

9(i)



Always give the min and max values on both axes.

Good practice to count the number of points plotted tallied with that given in question.

From the scatter diagram, as R increases, P increases at an increasing rate. Hence the relationship between P and R might not be well modelled by an equation of the form $P=aR+b$.

(ii)

For the model $P=aR+b$, $r=0.969$.
For the model $P=aR^2+b$, $r=0.993$.

Since the product moment correlation coefficient for the second model is closer to 1 than the first model, $P=aR^2+b$ is a better model.

Using GC,

$$P=\left(2.8457\times10^{-8}\right)R^2-0.28261 \quad (5\text{ s.f.})$$

$$P=\left(2.85\times10^{-8}\right)R^2-0.283 \quad (3\text{ s.f.})$$

Always give 5 s.f. answer before 3 s.f. answer.

(iii) Using equation from part (ii),

$$0.9=\left(2.8457\times10^{-8}\right)R^{2}-0.28261$$

Always use 5 s.f. answer for working.

$$R=6440 \quad (3\text{ s.f.})(\text{since } R \text{ is positive})$$

Therefore, when the power is 0.9 watts, an estimate for the speed of the fan is 6440 revolutions per minute.

Answer the question.

Since $P=0.9$ lies within the data range for P and $r=0.993$ is sufficiently close to 1, the estimate is reliable.

Need to fulfil **both** conditions before we can conclude that it is reliable.

(iv) Using equation from part (ii),

$$P=\left(2.8457\times10^{-8}\right)(3300)^{2}-0.28261$$

Always use 5 s.f. answer for working.

$$P=0.0273 \quad (3\text{ s.f.})$$

Therefore, when the the speed of the fan is 3300 revolutions per minute, an estimate for the power used is 0.0274 watts.

Answer the question.

Since $R=3300$ does not lie within the data range for R, the estimate is not reliable.

Need to fulfil **both** conditions before we can conclude that it is reliable. In this case, it failed one of the conditions, thus, the estimate is not reliable,

(v) 1 revolution per sec = 60 revolutions per min.
Thus, given R revolutions per sec, it is written as $60R$ revolutions per min.

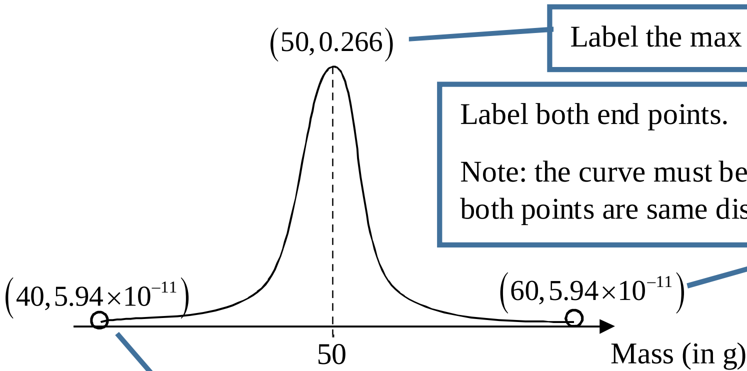
$$P=\left(2.8457\times10^{-8}\right)(60R)^{2}-0.28261$$

Always use 5 s.f. answer for working.

$$P=\left(1.0245\times10^{-4}\right)R^{2}-0.28261 \quad (5\text{ s.f.})$$

$$P=\left(1.03\times10^{-4}\right)R^{2}-0.283 \quad (3\text{ s.f.})$$

10(i)

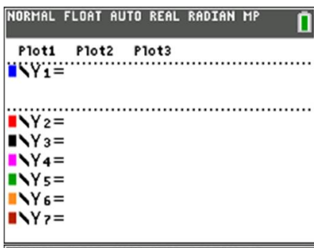


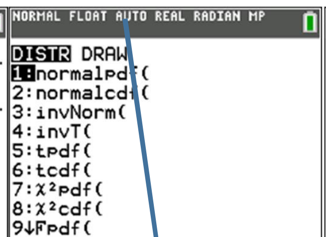
Label the max point

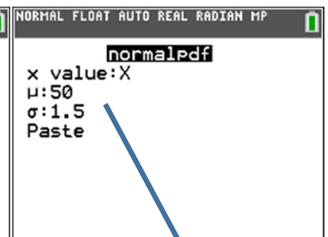
Label both end points.

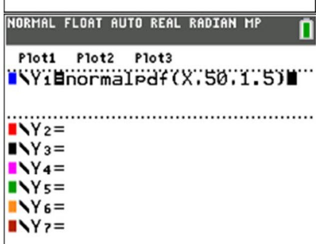
Note: the curve must be symmetrical and both points are same distance from the mean.

The endpoints must be **real close** to the axis as their y –values are almost zero



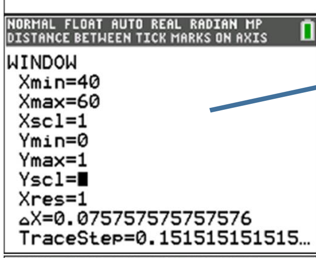






Use **normalpdf** to sketch the normal curve.

Key in the values as given. Only the x-values leave as x to sketch the



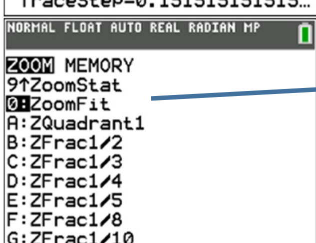
Key in

Xmin = 40 (sketching from 40 to 60)

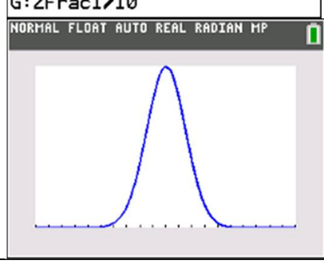
Xmax = 60 (sketching from 40 to 60)

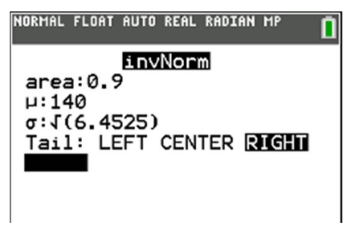
Ymin = 0

Ymax =1



Use ZoomFit to get the best screen output.



(ii)	Let L be the mass of a randomly chosen light bulb (in grams). $L \sim \mathrm{N}\left(50, 1.5^2\right)$ $\mathrm{P}\left(L < 50.4\right) = 0.605 \quad (3 \text{ s.f.})$
(iii)	Let B be the mass of a randomly chosen empty box (in grams). $B \sim \mathrm{N}\left(75, 2^2\right)$ $\mathrm{E}\left(B_1 + B_2 + B_3 + B_4\right) = 4(75) = 300$ $\mathrm{Var}\left(B_1 + B_2 + B_3 + B_4\right) = 4\left(2^2\right) = 16$ $T = B_1 + B_2 + B_3 + B_4 \sim \mathrm{N}\left(300, 4^2\right)$ $\mathrm{P}\left(T > 297\right) = 0.773 \quad (3 \text{ s.f.})$ 4 different boxes so total mass is $B_1 + B_2 + B_3 + B_4$, not $4B$.
(iv)	$\mathrm{E}\left(L + B\right) = 50 + 75 = 125$ $\mathrm{Var}\left(L + B\right) = 1.5^2 + 2^2 = 6.25$ $L + B \sim \mathrm{N}\left(125, 2.5^2\right)$ $\mathrm{P}\left(124.9 < L + B < 125.7\right) = 0.126 \quad (3 \text{ s.f.})$
(v)	Let W be the mass of the padding for a randomly chosen bulb (in g). $W = 0.3L$ $\mathrm{E}\left(W\right) = 0.3(50) = 15$ $\mathrm{Var}\left(W\right) = 0.3^2\left(1.5^2\right) = 0.2025$ $W \sim \mathrm{N}\left(15, 0.2025\right)$ $\mathrm{E}\left(L + W + B\right) = 50 + 15 + 75 = 140$ $\mathrm{Var}\left(L + W + B\right) = 1.5^2 + 0.2025 + 2^2 = 6.4525$ $L + W + B \sim \mathrm{N}\left(140, 6.4525\right)$ $\mathrm{P}\left(L + W + B > k\right) = 0.9$ $\therefore k = 136.74 = 137 \quad (3 \text{ s.f.})$ Apply Invnorm. 
(vi)	Let $X = L + W + B$. $\mathrm{E}\left(X_1 + X_2 + X_3 + X_4\right) = 4(140) = 560$ $\mathrm{Var}\left(X_1 + X_2 + X_3 + X_4\right) = 4(7.8025) = 25.81$ $Y = X_1 + X_2 + X_3 + X_4 \sim \mathrm{N}\left(560, 25.81\right)$ $\mathrm{P}\left(Y > 565\right) = 0.16251 = 0.163 \quad (3 \text{ s.f.})$